UNIT I – FUNDAMENTALS AND MATHEMATICAL FOUNDATIONS OF CRYPTOGRAPHY

Important Questions with Answers and Practice Problems – Based Only on the Provided PDF

PART A – IMPORTANT QUESTIONS FOR EXAMINATION

1. Define confidentiality, integrity and availability.

Confidentiality: Protecting information from unauthorized access or disclosure. Only authorized users or entities should access confidential information.
Integrity: Ensuring that information is not modified, deleted, inserted or altered by an unauthorized entity.
Availability: Ensuring that information and services are accessible to authorized users whenever required.

2. Explain the CIA Triad with suitable examples.

The CIA Triad consists of Confidentiality, Integrity and Availability.
Confidentiality: Prevents unauthorized disclosure. Example: Passwords should not be disclosed to unauthorized persons.
Integrity: Prevents unauthorized modification. Example: Changing “Transfer Rs. 5,000” to “Transfer Rs. 50,000” compromises integrity.
Availability: Ensures authorized access when needed. Example: An examination portal must remain accessible during registration.

3. Explain snooping and traffic analysis.

Snooping: Unauthorized access to or interception of data. An attacker attempts to observe or obtain information not intended for them.
Traffic analysis: Monitoring communication patterns even when message contents are protected. The attacker may study sender, receiver, frequency, timing and size of messages.

4. Explain modification, masquerading, replay and repudiation.

Modification: An attacker changes legitimate information after intercepting or accessing it.
Masquerading: Also called spoofing; an attacker pretends to be another legitimate entity.
Replaying: An attacker captures a valid message and retransmits it later.
Repudiation: One communicating party later denies having performed an action or having sent/received a message.

5. What is a Denial-of-Service attack?

A Denial-of-Service (DoS) attack attempts to make a system or service unavailable to legitimate users. An attacker may overload a server with requests or otherwise consume its resources.

6. Differentiate passive and active attacks.

Passive attack: Main objective is to obtain information; data is normally not changed; main concern is confidentiality. Examples: snooping and traffic analysis. Detection is often difficult.
Active attack: Main objective is to modify or disrupt; data is changed or system operation is affected; main concern is integrity and availability. Examples: modification, masquerading, replay, repudiation and DoS. Detection is usually comparatively easier.

7. List and explain the five security services.

1. **Data Confidentiality:** Protects information against unauthorized disclosure.
2. **Data Integrity:** Protects information against unauthorized modification, insertion, deletion and replay.
3. **Authentication:** Establishes confidence about the identity of an entity or the origin of data.
4. **Nonrepudiation:** Prevents a sender or receiver from later denying a communication or transaction.
5. **Access Control:** Protects resources against unauthorized access.

8. List and explain the security mechanisms.

Security mechanisms include: Encipherment, Data integrity mechanism, Digital signature, Authentication exchange, Traffic padding, Routing control, Notarization and Access control.
A security mechanism is a method used to implement a security service.

9. Differentiate security services and security mechanisms.

A **security service** describes **WHAT** protection is required. A **security mechanism** describes **HOW** that protection is implemented.

10. Explain symmetric-key and asymmetric-key cryptography.

Symmetric-key encipherment: Uses the same secret key for encryption and decryption. The communicating parties must possess or securely establish the shared secret key.
Asymmetric-key encipherment: Uses two related keys: a public key and a private key. For confidential communication to Bob, Alice can encrypt using Bob's public key and Bob can decrypt using his private key.

11. Differentiate cryptography and steganography.

Cryptography: Hides or transforms the meaning of a message and transforms plaintext into ciphertext.
Steganography: Hides the existence of a message and embeds information inside a cover such as an image, audio or video.

12. What is hashing?

Hashing converts a variable-length message into a fixed-length message digest. The digest can be used as a check value for data integrity.

13. Define divisibility and explain its notation.

If $a = q \times n$ for some integer q, then n divides a. It is written as $n \mid a$.
$n \mid a \Leftrightarrow a = q \times n$.

14. Calculate GCD using the Euclidean algorithm.

The Euclidean algorithm is an efficient method for calculating GCD and uses gcd(a, b) = gcd(b, r).
Example: $36 = 10 \times 3 + 6$; $10 = 6 \times 1 + 4$; $6 = 4 \times 1 + 2$; $4 = 2 \times 2 + 0$. Therefore, gcd(36, 10) = 2.

15. Explain the Extended Euclidean Algorithm.

The Extended Euclidean Algorithm calculates the GCD and also finds integers s and t satisfying: $a \times s + b \times t = \gcd(a, b)$. It is particularly important for finding multiplicative inverses in modular arithmetic.

16. Solve a linear Diophantine equation.

A linear Diophantine equation is $a \times x + b \times y = c$ and requires integer solutions for x and y. Let $d = \gcd(a, b)$. If d does not divide c, there is no integer solution. If d divides c, integer solutions exist and there are infinitely many solutions.

17. Define modular arithmetic and explain the modulo operator.

In modular arithmetic, we are mainly interested in the remainder obtained after division: $a \bmod n = r$, where $0 \leq r < n$. Here n is the modulus and r is the residue.

18. Define Z■ and residue class.

Z■ = {0, 1, 2, ..., n – 1}. A residue class [a] modulo n contains all integers that are congruent to a modulo n.

19. Differentiate equality and congruence.

Equality means two quantities are exactly the same. Congruence means they belong to the same residue class under a specified modulus. Example: $5 = 5$, but $5 \equiv 15 \pmod{10}$.

20. Explain additive and multiplicative inverses.

Additive inverse: a and b are additive inverses in Z■ if $a + b \equiv 0 \pmod{n}$.
Multiplicative inverse: a and b are multiplicative inverses modulo n if $a \times b \equiv 1 \pmod{n}$.

21. State and explain the condition for a multiplicative inverse.

a■$^{-1}$ exists modulo n $\Leftrightarrow \gcd(a, n) = 1$. Therefore, a and n must be relatively prime.

22. Find a modular inverse using the Extended Euclidean Algorithm.

If the Extended Euclidean Algorithm gives $s \times n + t \times b = 1$, then $t \times b \equiv 1 \pmod{n}$. Therefore, b■$^{-1} \equiv t \pmod{n}$.

23. Explain matrix multiplication and its dimension rule.

If A is an l × m matrix and B is an m × p matrix, then AB is an l × p matrix. Important rule: the number of columns of the first matrix must equal the number of rows of the second matrix.

24. Find the determinant of a 2 × 2 matrix.

For A = [a b; c d], det(A) = a × d – b × c. The determinant is defined only for square matrices.

25. Explain residue matrices and the condition for their inverse.

A residue matrix is a matrix whose elements belong to Z■. Matrix operations are performed using modular arithmetic. A residue matrix A is invertible modulo n if gcd(det(A), n) = 1.

26. Solve ax ≡ b (mod n).

Let d = gcd(a, n). If d does not divide b, there is no solution. If d divides b, there are exactly d solutions modulo n. General solution: x = x■ + k × (n/d), where k = 0, 1, ..., d – 1.

27. Explain how a system of linear congruences can be represented using matrices.

A set of linear congruences with the same modulus can be represented as A X ≡ B (mod n). If A has a multiplicative inverse modulo n, then X ≡ A■¹B (mod n). Here A is the coefficient matrix, X is the variable column matrix and B is the right-hand-side column matrix.

PART B – PRACTICE PROBLEMS WITH ANSWERS

1. Find the quotient and remainder when 527 is divided by 23.

527 = 22 × 23 + 21.
Quotient = 22, Remainder = 21.

2. Determine whether 17 divides 289.

289 = 17 × 17. Therefore, **17 divides 289.**

3(a). Find gcd(88, 220).

220 = 88 × 2 + 44; 88 = 44 × 2 + 0. Therefore, **gcd(88,220) = 44.**

3(b). Find gcd(300, 42).

300 = 42 × 7 + 6; 42 = 6 × 7 + 0. Therefore, **gcd(300,42) = 6.**

3(c). Find gcd(24, 320).

320 = 24 × 13 + 8; 24 = 8 × 3 + 0. Therefore, **gcd(24,320) = 8.**

3(d). Find gcd(401, 700).

700 = 401 × 1 + 299; 401 = 299 × 1 + 102; 299 = 102 × 2 + 95; 102 = 95 × 1 + 7; 95 = 7 × 13 + 4; 7 = 4 × 1 + 3; 4 = 3 × 1 + 1; 3 = 1 × 3 + 0. Therefore, **gcd(401,700) = 1.**

4. Use the Extended Euclidean Algorithm to find gcd(291,42) and the corresponding coefficients.

291 = 42 × 6 + 39; 42 = 39 × 1 + 3; 39 = 3 × 13 + 0. Therefore gcd = 3.
3 = 42 – 39 = 42 – (291 – 42 × 6) = 42 × 7 – 291.
Thus, **291(–1) + 42(7) = 3.** Coefficients: –1 and 7.

5(a). Find 22 mod 7.

22 mod 7 = 1.

5(b). Find 140 mod 10.

140 mod 10 = 0.

5(c). Find –78 mod 13.

–78 is exactly divisible by 13. Therefore, –**78 mod 13 = 0.**

5(d). Find 0 mod 15.

0 mod 15 = 0.

6. Calculate (273 + 147) mod 10.

273 + 147 = 420. Therefore, **420 mod 10 = 0.**

7. Calculate (125 × 45) mod 10.

125 × 45 = 5625. Therefore, **5625 mod 10 = 5.**

8. Find the additive inverse of every element in Z■■.

In Z■■, additive inverse pairs are: (0,0), (1,19), (2,18), (3,17), (4,16), (5,15), (6,14), (7,13), (8,12), (9,11), (10,10).

9. Find the multiplicative inverse pairs in Z■■.

The elements having multiplicative inverses are those relatively prime to 20: {1,3,7,9,11,13,17,19}. Inverse pairs are: (1,1), (3,7), (9,9), (11,11), (13,17), (19,19).

10. Find the inverse of 7 modulo 18.

7 × 13 = 91 and 91 ≡ 1 (mod 18). Therefore, **7■^{1} ≡ 13 (mod 18).**

11. Find the inverse of 11 modulo 26.

As given in the PDF: 11 × 19 = 209 ≡ 1 (mod 26). Therefore, **11■$^{-1}$ ≡ 19 (mod 26).**

12. Solve 3x ≡ 4 (mod 5).

3■$^{-1}$ ≡ 2 (mod 5). Therefore x ≡ 4 × 2 ≡ 8 ≡ **3 (mod 5).**

13. Solve 4x ≡ 4 (mod 6).

gcd(4,6) = 2 and 2 divides 4, so there are two solutions. Divide by 2: 2x ≡ 2 (mod 3). Thus x ≡ 1 (mod 3). Solutions modulo 6: **x = 1, 4.**

14. Solve 256x ≡ 442 (mod 60).

Reduce: 16x ≡ 22 (mod 60). gcd(16,60) = 4. Since 4 does not divide 22, **there is no solution.**

15. Solve 3x + 5 ≡ 4 (mod 5).

3x ≡ –1 ≡ 4 (mod 5). Since 3■$^{-1}$ ≡ 2 (mod 5), x ≡ 4 × 2 ≡ 8 ≡ **3 (mod 5).**

16. Find the determinant of a given 2 × 2 matrix.

The PDF does not provide a specific matrix in the practice problem. For A = [a b; c d], the determinant is **det(A) = ad** – **bc.**

17. Determine whether a residue matrix modulo 10 is invertible.

The PDF does not provide a specific matrix. A residue matrix A modulo 10 is invertible if **gcd(det(A),10) = 1.**

18. Solve a given system of two or three linear congruences using the matrix method.

The practice problem does not provide a specific system. The PDF states: A X ≡ B (mod n). If A has an inverse, then **X ≡ A■$^{-1}$B (mod n).**
The textbook example in the PDF has solution x ≡ 15, y ≡ 4 and z ≡ 14 (mod 16).